Differentiation

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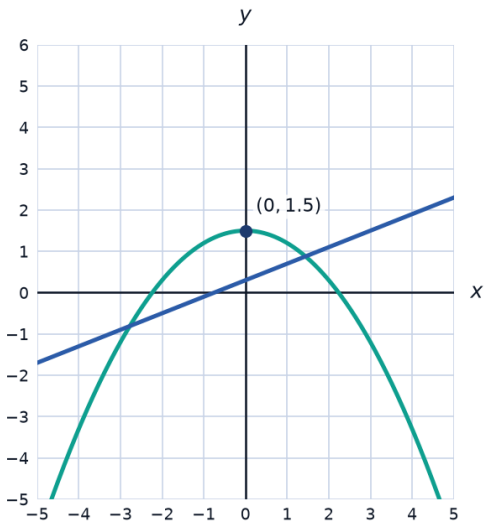
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Implicit differentiation**

Find $\frac{dy}{dx}$ given $x^3 + y^3 = 6xy$, and evaluate it at the point $(3, 3)$.

2 **[Foundational]** • **Target: Implicit differentiation**

Find $\frac{dy}{dx}$ given $\sin(xy) = x$, using implicit differentiation.



3 **[Foundational]** • **Target: Parametric differentiation**

A curve is defined parametrically by $x = t^2$, $y = 2t^3$. Find $\frac{dy}{dx}$ in terms of t.

4 **[Foundational]** • **Target: Parametric differentiation**

A curve is defined by $x = 3\cos t$, $y = 3\sin t$. Find $\frac{dy}{dx}$ in terms of t, and find the gradient at $t = \frac{\pi}{4}$.

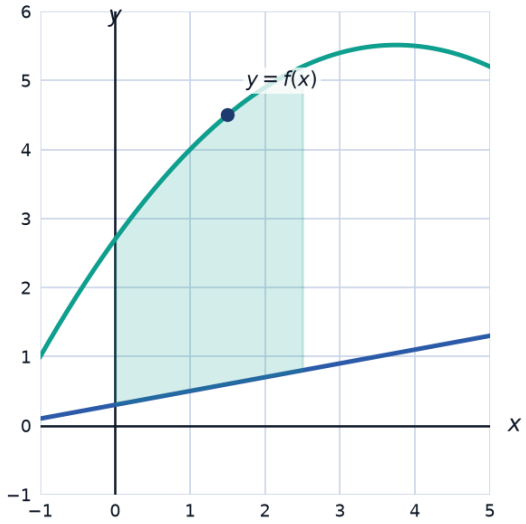
5 **[Foundational]** • **Target: Related rates**

A spherical balloon is inflated so its volume increases at a constant rate of 100 cm^3/s. Find the rate of increase of the radius when $r = 5$ cm, using $V = \frac{4}{3}\pi r^3$, to three decimal places.

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6 [Foundational] • Target: Related rates

The area of a circular ripple increases at 8π cm^2/s. Find the rate of change of the radius when $r = 4$ cm, using $A = \pi r^2$.



7 [Foundational] • Target: Related rates

A cone-shaped tank (vertex down) has height 10 cm and top radius 4 cm, so at water depth h the radius is $r = \frac{2h}{5}$. Water is poured in at 12 cm^3/s. Find the rate of rise of the water level when $h = 5$ cm, using $V = \frac{1}{3}\pi r^2 h$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Related rates

The radius of a circular oil slick increases at a constant rate of 0.5 m/min. Find the rate of increase of its area when $r = 20$ m.

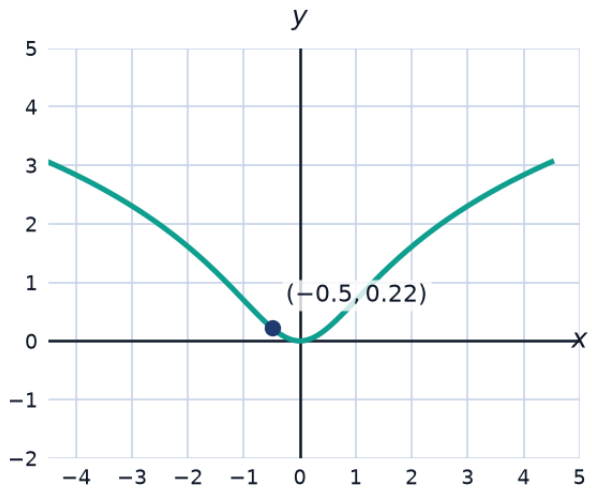
9 [Intermediate] • Target: Applying the rules: tangents and second derivatives

Find the equation of the tangent to $y = xe^{-x}$ at $x = 1$.

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10 [Intermediate] • Target: Applying the rules: tangents and second derivatives

Find $\frac{d^2y}{dx^2}$ for $y = \ln(x^2 + 1)$, and evaluate it at $x = 1$.



11 [Intermediate] • Target: A well-designed word problem: a ladder sliding down a wall

This question has three parts. A ladder 5 m long leans against a vertical wall, its base on horizontal ground. The base slides away from the wall at a constant rate of 0.4 m/s. (a) Let x be the distance of the base from the wall and y the height of the top of the ladder. Write the relationship between x and y, and differentiate implicitly with respect to time t to relate $\frac{dx}{dt}$ and $\frac{dy}{dt}$. (b) Find the rate at which the top of the ladder is sliding down the wall when the base is 3 m from the wall. (c) Using $x = 5\cos\theta$, where θ is the angle between the ladder and the ground, find the rate at which θ is changing at that instant.

12 [Intermediate] • Target: The quotient rule

Level 2 · MediumFind the derivative $\frac{dy}{dx}$ of $y = x^{\sin x}$ for $x > 0$ using logarithmic differentiation.

13 [Intermediate] • Target: Implicit differentiation

Level 1 · FoundationsFind the derivative of $y = \sqrt{x^4 + 3x^2 + 7}$ using the Chain Rule, and compute $y'(1)$.

14 [Intermediate] • Target: Parametric differentiation

Level 3 · Advanced A particle moves along a curve defined parametrically by $x = t^2 + 1$ and $y = t^3 - 3t$. Find the value of $\frac{d^2y}{dx^2}$ at $t = 2$ and determine whether the curve is concave up or concave down at this point.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

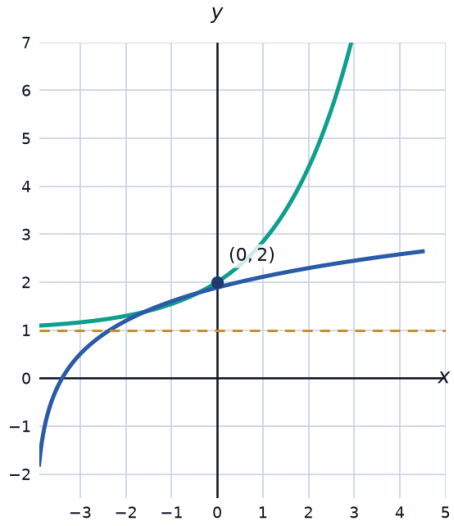
15 [Challenging] • Target: Related rates • Cross-Lesson: 5.01 Integration

Level 3 · AdvancedEvaluate the indeterminate limit using repeated applications of L'Hôpital's Rule with complete hypothesis verification: $\lim_{x \to 0} \frac{x - \arctan x}{x^3}$.

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16 **[Challenging] • Target: Differentiating trigonometric, exponential and logarithmic composites • Cross-Lesson: 6.01 Sequences & Series**

Solve the logarithmic equation $\log_3(x+6) + \log_3(x) = 3$ and eliminate any extraneous solutions.



17 **[Challenging] • Target: Applying the rules: tangents and second derivatives • Cross-Lesson: 7.01 Vectors**

A curve has equation $y = 2x^3 - 9x^2 + 12x + 1$. (a) Find the coordinates of both stationary points. (b) Determine the nature of each stationary point using the second derivative $\frac{d^2y}{dx^2}$.

18 **[Challenging] • Target: A well-designed word problem: a ladder sliding down a wall • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division**

A radioactive substance follows the decay model $M(t) = M_0 e^{-kt}$. If 20% of the sample decays in 15 days, find the exact decay constant k and the substance's half-life in days to the nearest tenth.

19 **[Challenging] • Target: The chain rule • Cross-Lesson: 9.01 Complex Numbers**

[Exponential, Logarithmic & Logistic Dynamics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

20 **[Challenging] • Target: The product rule • Cross-Lesson: 1.01 Functions: Modulus & Inverse**

Level 3 · AdvancedEvaluate the limit analytically without using L'Hôpital's Rule: $\lim_{x\to 0} \frac{\sqrt{1+\tan x} - \sqrt{1+\sin x}}{x^3}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The chain rule

Related rates: a 10-m ladder slides down a wall. When the base is 6 m from the wall and moving away at 2 m/s, the top is moving at:

a. $-\frac{3}{2}$ m/s

b. $-\frac{3}{4}$ m/s

c. $\frac{3}{2}$ m/s

d. $\frac{4}{3}$ m/s

22 [Foundational] • Target: The product rule

A ladder 10 m long leans against a wall; the base slides away at 1 m/s. How fast is the top sliding down when the base is 6 m from the wall?

a. $\frac{3}{4}$

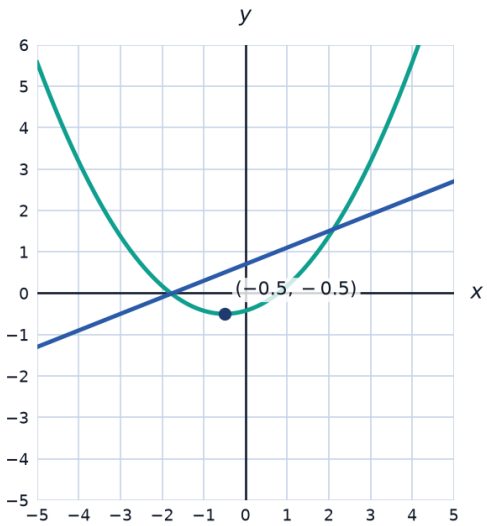
b. $-\frac{3}{8}$

c. $-\frac{3}{4}$

d. $-\frac{3}{2}$

23 [Foundational] • Target: The quotient rule

Related rates: ladder 10 m against wall. Base slides at 1 m/s when base is 6 m. Rate of top:



a. $-3/4$ m/s

b. $3/4$ m/s

c. -1 m/s

d. $-4/3$ m/s

24 [Foundational] • Target: Implicit differentiation

Solve $x^2 - 2x + 17 = 0$. Find the product of the roots.

a. 16

b. 2

c. 17

d. 18

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25 [Foundational] • Target: Parametric differentiation

A ladder 13 m long leans against a wall. Base slides at 2 m/s. Rate of top falling when base is 5 m from wall:

a. $-\frac{5}{6}$ m/s

b. $-\frac{5}{12}$ m/s

c. $-\frac{2}{5}$ m/s

d. $-\frac{1}{2}$ m/s

26 [Foundational] • Target: Related rates

A 10-ft ladder leans against a wall; the base is 6 ft from the wall and slides away at $\frac{dx}{dt} = 1$ ft/s. Find $\frac{dy}{dt}$, the rate the top slides down (height $y = 8$ ft).

a. -1 ft/s

b. $-\frac{3}{4}$ ft/s

c. $-\frac{1}{4}$ ft/s

d. $-\frac{1}{2}$ ft/s

27 [Foundational] • Target: Differentiating trigonometric, exponential and logarithmic composites

Implicit differentiation: $\frac{d^2y}{dx^2}$ at $(1, 1)$ for $x^2 + y^2 = 2$:

a. -1

b. -2

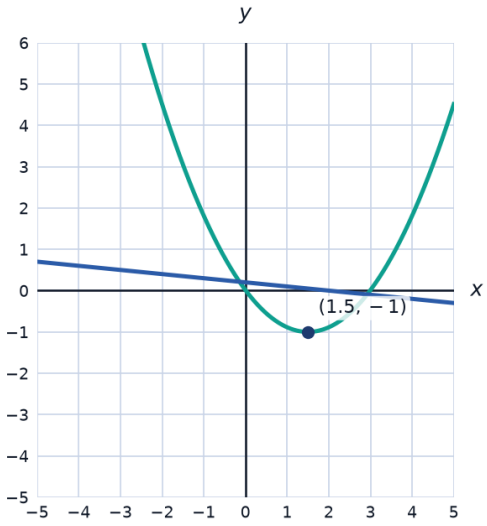
c. 0

d. 1

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28 [Foundational] • Target: Applying the rules: tangents and second derivatives

Related rates: water drains from a conical tank (height $H = 10$m, radius $R = 5$m) at 2 m³/min. When depth $h = 4$ m, $dh/dt =$



a. $-\frac{2}{4\pi}$ m/min

b. $-\frac{1}{2\pi}$ m/min

c. $-\frac{2}{\pi}$ m/min

d. $-\frac{8}{\pi}$ m/min

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: a ladder sliding down a wall

Related rates: if $A = r^2\pi$, then $\frac{dA}{dt} =$

a. $2\pi r$

b. $2\pi r\frac{dr}{dt}$

c. $\pi r^2\frac{dr}{dt}$

d. $2\pi\frac{dr}{dt}$

30 [Intermediate] • Target: The chain rule

Related rates: a balloon radius expands at 2 cm/s. Rate of volume change when $r = 5$ cm is:

a. 40π cm³/s

b. 100π cm³/s

c. 200π cm³/s

d. 300π cm³/s

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31 **[Intermediate]** • **Target: The product rule**

Logarithmic differentiation: if $y = \frac{x^2\sqrt{x+1}}{e^x}$, then $y' =$

a. $y\left(\frac{2}{x} + \frac{1}{2(x+1)} - 1\right)$

b. $y\left(\frac{2}{x} - \frac{1}{2(x+1)} - 1\right)$

c. $y\left(\frac{2}{x} + \frac{1}{2(x+1)} + 1\right)$

d. $y\left(2x + \frac{1}{2} - 1\right)$

32 **[Intermediate]** • **Target: The quotient rule**

Logarithmic differentiation: $y = x^x$. $y' =$

a. $x^x(1 + \ln x)$

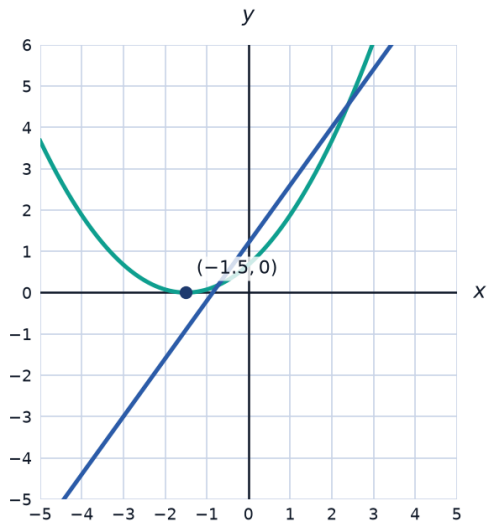
b. $x \cdot x^{x-1}$

c. $x^x \ln x$

d. e^x

33 **[Intermediate]** • **Target: Implicit differentiation**

Chain rule with two paths: $z = f(x, y)$, $x = r\cos\theta$, $y = r\sin\theta$. $\partial z/\partial r =$



a. $f_x\cos\theta + f_y\sin\theta$

b. f_r

c. $f_x r + f_y r$

d. $f_x\sin\theta - f_y\cos\theta$

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34 [Intermediate] • Target: Parametric differentiation

Implicit differentiation of $xy = 5$: $\frac{dy}{dx} =$

a. $\frac{5}{x}$

b. $\frac{y}{x}$

c. $-\frac{y}{x}$

d. $\frac{x}{y}$

35 [Intermediate] • Target: Related rates

Using the Product Rule, $\frac{d}{dx}[(3x+1)(x^2-2)] =$

a. $3(x^2-2)+2x(3x+1)$

b. $3x^2-6+6x^2+2x$

c. $9x^2+2x-6$

d. all are equivalent

36 [Intermediate] • Target: Differentiating trigonometric, exponential and logarithmic composites

Related rates: sphere volume $V = (4/3)\pi r^3$. If $dr/dt = 2$ cm/s at $r = 3$ cm, $dV/dt =$

a. 72π cm³/s

b. 36π cm³/s

c. 12π cm³/s

d. 24π cm³/s

37 [Intermediate] • Target: Applying the rules: tangents and second derivatives

Implicit differentiation: $F(x, y) = 0$. Then $dy/dx =$

a. F_y/F_x

b. $-F_x/F_y$

c. F_x/F_y

d. $-F_y/F_x$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: a ladder sliding down a wall • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

A ladder 10 m long leans against a wall. If bottom slides out at 1 m/s, this is a:

a. linear rate

b. related rates problem

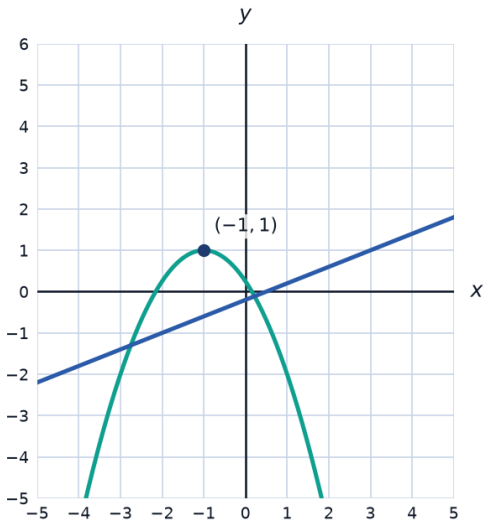
c. optimisation

d. linearisation

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39 [Challenging] • Target: The chain rule • Cross-Lesson: 9.01 Complex Numbers

A geometric sequence has first term $a = 2$ and common ratio $r = 2$. Find the 5th term.



a. 64

b. -32

c. 34

d. 32

40 [Challenging] • Target: The product rule • Cross-Lesson: 1.01 Functions: Modulus & Inverse

A line has equation $\mathbf{r} = (-2, 7, 12) + t(7, 4, 1)$. Find the x-coordinate of the point where $t = -5$.

a. -30

b. 7

c. -13

d. -37

41 [Challenging] • Target: The quotient rule • Cross-Lesson: 10.01 Numerical Methods

For the locus $|z - (3 + 4i)| = 4$, find the maximum value of $|z|$.

a. 1

b. 5

c. 9

d. 4

42 [Challenging] • Target: Implicit differentiation • Cross-Lesson: 11.01 Differential Equations

A quantity decays according to $\frac{dy}{dt} = -ky$. If the quantity halves every 1 year, find k.

a. $\log(2) + 2$

b. $\log(2)$

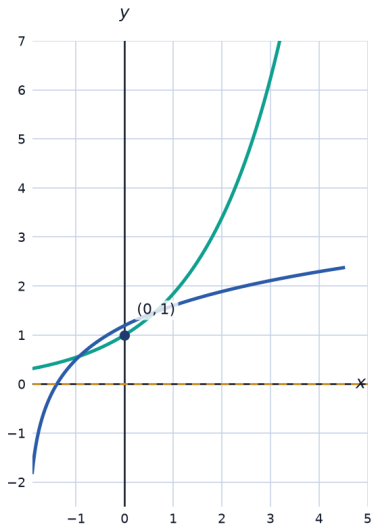
c. $\log(3)$

d. $\log(2) + 1$

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43 [Challenging] • Target: Parametric differentiation • Cross-Lesson: 12.01 Poisson Distribution

A quantity decays according to $\frac{dy}{dt} = -ky$. If the quantity halves every 2 years, find k.



a. $2\log(2)$

b. $\frac{\log(3)}{2}$

c. $\frac{\log(2)}{2} + 1$

d. $\frac{\log(2)}{2}$

44 [Challenging] • Target: Related rates • Cross-Lesson: 13.01 Continuous Random Variables

A line has equation $\mathbf{r} = (7, -7, 0) + t(2, -3, -4)$. Find the z-coordinate of the point where $t = 2$.

a. -13

b. 11

c. -12

d. -8

45 [Challenging] • Target: Differentiating trigonometric, exponential and logarithmic composites • Cross-Lesson: 14.01 Sampling & Estimation

A line has equation $\mathbf{r} = (2, -6, -2) + t(-4, -1, -4)$. Find the y-coordinate of the point where $t = -2$

.

a. -5

b. 10

c. -4

d. 6

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